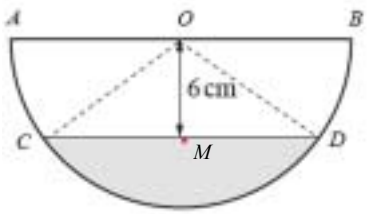
| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 3(b)(iii) | Correct explanation which states the median is 22 and correct reference to 100 or 101<br>e.g.<br>• Median is 22 which is 50% of the people and 101 is more than 50% oe<br>• The median is 22 which is the 100th number (accept 100.5th number) | **1** | |
| 4(a)(i) | 22 620 cao | **3** | **B2** for 7200π or 22 608 to 22 629<br>or **M1** for $\frac{1}{2}\times\pi\times12^2$[×figs 1] oe |
| 4(a)(ii) | 8840 or 8850 or 8836 to 8850.<br>  <br> | **5** | **M1** for cos $COM = \frac{6}{12}$ oe<br>or sin $AOC = \frac{6}{12}$ oe<br>**M1** for $\left(\frac{their\ COD}{360}\times\pi\times12^2\right)$ oe<br>**M1** for $\left(\frac{1}{2}\times12^2\times\sin(their\ COD)\right)$ oe<br>**M1dep** for (*their* area of sector *COD*–*their* area of triangle *COD*) ×100 **dep** on at least M1M1 oe |
| 4(b) | 647 or 646.8 | **3** | **M2** for $2.2 = \frac{m}{42\times35\times0.2}$ oe<br>or **M1** for [vol of stone =] 42×35×0.2 oe<br>If 0 scored **SC1** for answer figs 647 or figs 6468 |
| 4(c) | 46.1 or 46.12 to 46.14 | **4** | **M3** for $\tan = \frac{15}{\sqrt{8^2+12^2}}$ oe<br>or **M2** for $8^2 + 12^2$ oe or $8^2 + 12^2 + 15^2$ oe<br>or **M1** for identifying the angle *GAC* |
| 5(a) | $125x^9$ final answer | **2** | **B1** for answer $125x^k$ or $mx^9$<br>or for correct answer seen then spoilt |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 5(b) | $6^{n-2}$ oe final answer | **2** | **B1** for answer of form $6^k$ oe<br>or answer of the form $\left(\frac{1}{6}\right)^{-k}$ oe<br>or for correct answer seen |
| 5(c) | $3x^3 + 2x^2 - 37x + 12$ final answer | **3** | **B2** for correct expansion of three brackets unsimplified or for simplified four-term expression of correct form with 3 terms correct<br><br>or **B1** for correct expansion of two brackets with at least 3 terms out of 4 correct |
| 5(d)(i) | eliminates the fraction correctly<br>eg $(3x + 5)(x - 2) + 7 = x(x - 2)$ | **M1** | |
| | $3x^2 + 5x - 6x - 10 + 7 = x^2 - 2x$ oe | **B2** | **B1** for $3x^2 + 5x - 6x - 10$ [ + 7] oe seen with at least 3 terms correct |
| | leading to $2x^2 + x - 3 = 0$ | **A1** | **dep** on M1 B2<br>with no errors or omissions |
| 5(d)(ii) | $(2x+3)(x-1)$ | **M2** | or **M1** for $(2x + a)(x + b)$ where $ab = -3$<br>or $2b + a = [+]1$<br>or for partial factors<br>$2x(x - 1) + 3(x - 1)$<br>or $x(2x + 3) - [1](2x + 3)$ |
| | $-1.5$ oe and $+1$ | **B1** | |
| 5(e) | [TSA cylinder =] $2\pi x^2 + 2\pi x \times 3x$ | **M1** | |
| | [TSA hemisphere=] $\pi(5y)^2 + \frac{4\pi(5y)^2}{2}$ | **M1** | |
| | Leading to<br>$2\pi x^2 + 6\pi x^2 = 50\pi y^2 + 25\pi y^2$ oe | **M1** | dep **M1M1** |
| | $x^2 = \frac{75y^2}{8}$ | **A1** | dep on **M1M1M1** |
| 6(a) | $\sqrt{10.4^2 + 6.5^2 - 2 \times 10.4 \times 6.5 \times \cos 64}$ | **M2** | **M1** for $10.4^2 + 6.5^2 - 2 \times 10.4 \times 6.5 \times \cos 64$<br>**A1** for 91.1 to 91.2 |
| | 9.546 to 9.547 | **A1** | |